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| **1.5 The functions of mechanical devices used to produce different sorts of movements, including the changing of magnitude and the direction of forces** | The performance, principles, applications and the influence on the design of products of the following. |
| | 1.5.1 Types of movement:<br>a linear<br>b reciprocation<br>c rotary<br>d oscillation. |
| | 1.5.2 Classification of levers:<br>a class 1, 2 and 3<br>b calculations related to mechanical advantage (MA), velocity ratio (VR), load, effort and efficiency. |
| | 1.5.3 Linkages:<br>a bell crank<br>b reverse motion linkages. |
| | 1.5.4 Cams:<br>a pear shaped<br>b eccentric (circular)<br>c drop (snail). |
| | 1.5.5 Followers:<br>a roller<br>b knife<br>c flat followers. |
| | 1.5.6 Pulleys and belts:<br>a V-belt<br>b velocity ratio (VR)<br>c input and output speeds. |

| Key idea | What students need to learn: |
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| | 1.5.7 Cranks and sliders. |
| | 1.5.8 Gear types:<br>a simple and compound gear train<br>b idler gear<br>c revolutions per minute (RPM) calculations<br>d bevel gears<br>e rack and pinion. |